

ECCT 2026: Responses to the reviewers' Queries/Comments

Paper ID: 482

paper Title: MANet: A Context-Aware Attention Network for Robust Zero-Day Ransomware Detection

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Reviewer#4

Concern # 1: Referencing is not sequential. "techniques [11,?]" Reference list is not correct.

Author(s) response: Thank you for highlighting this problem with our format. We realize that there were problems with the numbering because we have done many inserts during the revision process. They have all been correctly reordered according to their correct first appearance number in the manuscript. Nowadays, each citation is numbered using integers, such as [1], [2], [3].

Author(s) action in the paper:

Correctly reorder all 16 references according to their correct first appearance number.

- Confirm that the citation number matches the corresponding reference in the list.
- As opposed to conference papers citations guidelines

Concern # 2: Avoid 'et al.' in the reference list, write all authors' names

Author(s) response: As the reviewer correctly pointed out, the conventional format requires the inclusion of all names on the reference list, rather than "et al." Therefore, the references have all been made complete by adding all the names of the authors.

Author(s) action in the paper:

- All 16 references have been made complete by listing all the names of the authors.

Concern # 3: The architectural contribution is largely an adaptation rather than a fundamentally new method.

Author(s) response: In this respect, we humbly clarify how MANet's architecture innovates. In particular, the innovation of MANet lies in designing a domain-specific ransomware detection model

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as a feature interaction problem based on a specific modification of the Transformer's Multi-Headed Self-Attention mechanism. Some novel parts include:

1. A special Behavioural Embedding Layer, as opposed to general token embeddings, developed specially for 483 RFE-selected behavioural features
2. A deep classification MLP network calibrated for malware detection via gradual BatchNorm reduction (256 → 128 → 64)
3. The architectural evaluation takes into account F2-Score threshold calibration, whereas no attention-based ransomware detector exploits the mathematically justified SOC-friendly approach.
4. A Latent Attention Heatmap (16x16 attention heat mapping for behavioural embeddings).

Author(s) action in the paper:

- A comparative table, along with a new "Positioning of MANet" subsection, is added to Related Work.
- To highlight the domain-driven considerations taken into account in the proposed architecture, Section 3.3 has been revised.
- More content added to the Introduction: Three special architectural components that are only considered in ransomware detection by the MANet model include (i) behavioral embedding layer with LayerNorm as the means of converting tabular data into sequences; (ii) F2-calibrated decision boundaries due to the cost-asymmetric SOC problem; and (iii) the attention heatmap.

Reviewer#7

Concern # 1: Single dataset with only one zero-day family (Akira) — weak generalizability

Author(s) response: We acknowledge this limitation and have addressed it by **validating MANet on a second independent dataset:** the Ransomware Dataset 2024 (Mexwell, Kaggle), containing **21,752 samples across 27 malware families** including LockBit, Ryuk, WannaCry, REvil, Phobos, GandCrab, and Emotet — none of which appear in MLRan.

Metric	MLRan (Original Dataset)	Ransomware Dataset 2024
Accuracy	98.26%	98.80%
F1-Score	0.9817	0.9877
Brier Score	0.0161	0.0158
Zero-day Families	1 (Akira)	26 (all test families unseen during training)

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Author(s) action in the paper:

- "Cross-Dataset Validation on Ransomware Dataset 2024" has been added to Section 5.2.1.
- Results of "MLRan vs. Ransomware Dataset 2024" have been compared in Table 3.
- A reference for the Zenodo dataset ("cite{zenodo2024ransomware}") has been included in the text.
- Limitations section has been revised including cross-d

Concern # 2: Very high-performance metrics raise concerns about dataset bias or task simplicity.

Author(s) response: This is a valid concern. We have addressed it with **two-dataset validation** and **statistical significance testing**:

Author(s) action in the paper:

This is a valid concern. We have addressed it with two-dataset validation and statistical significance testing

Dataset Characteristic	MLRan	Ransomware Dataset 2024
Feature Count	483	18
Feature Type	RFE behavioral	PE headers + API + registry
Class Balance	1.09:1	1:1 (perfect)
Evaluation Protocol	Chronological (TESSERACT)	Random stratified (no family overlap)
MANet Accuracy	98.26%	98.80%

The consistent **98%+ accuracy** across different feature spaces, class distributions, and evaluation protocols indicates strong generalization ability of the model, suggesting that the performance is driven by genuine learning rather than dataset-specific biases or artifacts.